

What Drives SPECTRUM ANALYSERS



Dilin Anand is a senior assistant editor at EFM. He is B.Tech from University of Calicut, currently pursuing MBA from Christ University, Bengaluru

While incremental specification enhancements are taken as a given in any field driven by technology, the past year has seen some interesting advances on the side of improving test and measurement (T&M) productivity. We have seen support being integrated into the equipment for some of the newer standards, better automation features as well as a renewed focus on making T&M a much better experience for the engineer. This article takes a look at the trends for the latest line of signal and spectrum analysers.

Significant improvements in bandwidth and frequency

We are seeing some significant increments in specification enhancements. "Recently, spectrum analyser lines of equipment have improved their capabilities by moving forward from featuring frequency coverage of around 20GHz to enter the millimetre range of up to 50GHz. Thus their frequency range has been enhanced quite a bit," explains Vishal Gupta, senior applications specialist (RF/MW), Keysight Technologies. The modular PCI eXtensions for Instrumentation (PXI) form factor now features up to 26.5GHz of coverage, too.

Newer equipment come with higher analysis bandwidth. The UXA branded line from Keysight can handle up to 510MHz of real-time analysis bandwidth. This is a significant increase as the best bandwidth topped at 160MHz in 2014, adds Gupta.

"Using analysers for frequencies

above 6.0GHz is a big challenge for designers. The most exciting feature/function of the spectrum analyser is to make the single-side-band noise phase to be 100dBc/Hz at offset up to 10Hz, which is rather hard to achieve," explains Chandmal Goliya, director, Kusam Electrical Industries Ltd.

Equipment like Spectran V5 real-time spectrum analysers (RTSAs) from Aaronia AG allow continuous analysis and even real-time data streaming (for example, USB to hard disk).

"This allows all data of a desired high-frequency band to be recorded without any blind spot (for example, complete data traffic of a mobile phone tower). Real-time bandwidth can reach up to 200MHz (existing systems can reach only up to 110MHz). An ultra-fast direct-digital-synthesis sweep in S-area is possible even with 10GHz bandwidth," explains Naveen Sharma, technical manager, Scientific Mes-Technik Pvt Ltd.

Size reduction still a focus area

For some time now, there has been an emphasis on size reduction, which has continued to the point that we now have handheld instruments, explains Gupta.

Goliya reiterates, "We think the handheld spectrum analyser is a good application for the market that values portable and lightweight equipment for outdoor use and constantly changing workplace requirements."

Satish Thakare, chief technology officer, Scientech Technologies Pvt Ltd, feels that, "Traditional spectrum analyser instruments available in the handheld and bench-top form factors are still popular, but modular instruments are finding wider acceptance in the market due to cost, flexibility and scalability advantages."

Three clicks or less

Rufus Danesh, project engineer at Analog Arts, says, "What makes modern signal

Signal or Spectrum Analysers

Spectrum and signal analysers have come a long way from where these were a few years ago. Until recently, while most signal analysers came with basic spectrum analyser capabilities, these were unable to perform as well as a real spectrum analyser. However, things have changed now with the integration of a lot more features throughout equipment lines. "We do not call these spectrum analysers anymore, we now call them signal analysers," says Vishal Gupta, senior applications specialist (RF/MW), Keysight Technologies.